

OfflineMind

Private, offline document Q&A for Snapdragon® PCs

Snapdragon® AI Lab Build & Present Challenge | Arpit Bhardwaj

The problem

- Students and professionals keep sensitive PDFs: notes, contracts, reports, medical files
- Cloud AI tools need uploads, internet and subscriptions
- Poor connectivity and privacy worries block AI use for many people in India

Our solution

- Ask questions over your own documents, fully on your laptop
- Retrieval-augmented generation: embed, search, answer with source pages
- No internet, no data leaves the device, no per-use cost

How it works

- 1. Load PDFs and split into chunks
- 2. Embedding model (ONNX) builds a local vector index
- 3. Top matches feed a quantized local LLM
- 4. Answer shown with citations to the source page
- Runtime prefers the Snapdragon NPU via ONNX Runtime QNN, with CPU fallback

Why Snapdragon

- NPU runs models at low power, so long sessions do not drain the battery
- On-device inference means low latency and full privacy
- Models sourced from Qualcomm AI Hub and open-source platforms
- Benchmark plan: NPU vs CPU latency and tokens/sec on the target HP PC

Deployment and accessibility

- Install in a few commands: pip install, download model, run Streamlit
- Clear README, requirements file and model download script
- Works offline after first setup; simple browser-style UI

Impact and next steps

- Students: search notes and textbooks offline
- Professionals: query private files with zero data exposure
- Next: Indian-language support, voice input, OCR for scanned notes